Alireza Amjadi

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Education

Sharif University of Technology Tehran, Iran
B.Sc. in Mechanical Engineering, Dynamics, Control and Vibration track *Sep 2023 – Present | Expected graduation: Jul 2027*

- **GPA:** 18.69/20 **Rank:** 11th of 126 in the Mechanical Engineering cohort (top 9%)
- **GPA over the last two years:** 19.27/20 **Most recent semester:** 19.61/20
- **Key courses (out of 20):** Dynamics of Machinery (20), Measurement and Control Systems (19.6), Vibrations (19.3), Automatic Control (18.5), Dynamics (18.8), Engineering Mathematics (20), Differential Equations (19.8), Fundamentals of Electrical Engineering I and II (20, 20), Heat Transfer I (20).
- **Current courses (Fall 2026):** Musculoskeletal Biomechanics (graduate course, Prof. Farzam Farahmand), Robotics, Automatic Control Design, Automatic Control Laboratory.

Research Interests

Biomechatronics · Wearable Robotics and Exoskeletons · Active Orthoses and Assistive Devices · Rehabilitation Robotics · Musculoskeletal Modelling and Simulation · Gait Biomechanics · Human–Robot Interaction · Dynamics, Control and Vibration · Robotics and Mechatronics · Physics-Informed Machine Learning

Research Experience

B.Sc. Project: Active Ankle–Foot Orthoses for Post-Stroke Drop Foot Sharif University of Technology
Supervisor: Prof. Farzam Farahmand *Fall 2026 – Present*

- Active ankle–foot orthoses for stroke survivors who cannot lift the forefoot while walking (drop foot).
- Responsible for the musculoskeletal modelling and simulation of the human–orthosis system in OpenSim.
- Reviewing the literature on modelling active and passive assistive devices in OpenSim 4.6.

Technical Skills

- **Biomechanics & Dynamics:** OpenSim, Simscape Multibody, Simulink, multibody dynamics, control design.
- **Computational Mechanics:** ANSYS Fluent, ANSYS Mechanical, finite-volume methods, verification and validation.
- **Programming & Machine Learning:** Python (NumPy, SciPy, PyTorch), MATLAB, PINNs.
- **CAD & Mechanical Design:** SolidWorks, CATIA, FEA, fatigue design of gears, shafts and bearings.
- **Tools:** LaTeX, Git, EES.
- **Languages:** Persian (native), Azerbaijani Turkish (native), English (fluent), Turkish (intermediate).

Selected Academic Projects

Newest first. Each title links to its GitHub repository, with the code and a full English report.

Pump Mechanism and Flywheel: Multibody Dynamics *Spring 2026*
Dynamics of Machinery · Prof. Hassan Sayyaadi · Individual project

- Built a Simscape Multibody model of an oscillating-cylinder pump; peak torque about 16/48 N m at 100/300 rpm.
- Sized a flywheel in MATLAB and verified it in simulation: speed fluctuation cut to 5.71% (target: below 10%).

DC Motor Modelling and PI Speed Control *Spring 2026*
Fundamentals of Electrical Engineering II · Team project (3 students); contributed to the technical report

- Simulink model of a separately excited DC motor from its equations, matching the analytical steady state.
- P and PI speed controllers with armature-voltage saturation, tested under a 1.5 N m load.

Hybrid Liquid–Air Cooling of Power Electronics *Spring 2026*
Heat Transfer I · Dr. Mohamad Ali Bijarchi · Team project (2 students); wrote the solver and the full report

- Wrote a transient finite-volume conduction solver from scratch in MATLAB; explicit and implicit schemes agree.
- Grid and time-step independence shown (GCI $\approx$ 0.41%); agreement with ANSYS Mechanical within 0.57%.

Airfoil Flow with CFD, PINNs and a Source–Sink Method *Fall 2025*
Fluid Mechanics II · Dr. Mohamad Sadeq Karimi · Individual project

- Studied the cambered NACA 4421 and the symmetric NACA 16-006 airfoils with three methods.
- NACA 4421: Fluent (RANS) with a five-mesh grid study (C_L within 0.1%) and a Navier–Stokes PINN in PyTorch.
- NACA 16-006: potential-flow PINN, plus a source–sink method in MATLAB for five airfoil thicknesses.

Gearbox Gear Design and Final Layout *Fall 2025*
Machine Elements Design II · Prof. Mohammad Durali · Team project (2 students); led the calculations and wrote the report

- Designed helical and straight bevel gear stages for infinite life in bending and surface fatigue.
- Updated the shafts, bearings and housing for the final gear forces; FEA of the housing, drawings and cost estimate.

Mine Haulage Winch *Fall 2025*
Machine Elements Design II · Prof. Mohammad Durali · Team project (2 students); led the calculations and wrote the report

- Winch for a 5-tonne cart in a 25° tunnel: belts, wire rope, chain, drum, shaft, bearings and chassis.

Gearbox Shafts, Bearings and Housing *Fall 2025*
Machine Elements Design II · Prof. Mohammad Durali · Team project (2 students); led the calculations and wrote the report

- Fatigue design of shafts and keys for an 11 kW gearbox; SKF bearing selection; welded housing checked by FEA.

3D Physics-Informed Neural Network for Pipe Flow *Spring 2025*
Fluid Mechanics I · Dr. Mohamad Sadeq Karimi · Individual project

- Wrote a 3D Navier–Stokes PINN from scratch in NumPy, including its own automatic differentiation.
- Recovered the Hagen–Poiseuille solution within 0.2% and the entrance length within 4.6% of published data.

Fork-Mounted Jib Crane *Spring 2025*
Machine Elements Design I · Prof. Mohammad Durali · Team project (2 students); led the calculations and wrote the report

- Demountable crane for a 3-tonne forklift: beams, pins, welds and bolted joints, with a user catalogue.

Industrial Experience

Ahan Panjeh Industrial Group Tabriz, Iran
Industrial Intern, perforated sheet and expanded metal manufacturer (240 hours, graded Excellent) *Summer 2026*

- Set up CNC punch presses with the die-making team: power train, table feed, servo drives and encoders.
- Selected punches and dies for orders, checked tool-steel quality and estimated the shear force each order needs.
- Documented the machine's stored G/M-code programs and compared its control with the hydraulic press line.

Teaching Experience

Grader, Statics Sharif University of Technology
Instructor: Dr. Somaye Mohammadi *Spring 2026*

Instructor, Differential Equations Problem-Solving Workshop Sharif University of Technology
Organised by Mehvar Scientific Association, Department of Mechanical Engineering *Nov 2025*

- Led a 90-minute session of exercises, sample midterm problems and Q&A (own course grade: 19.8/20).

Leadership and Service

Peer Mentor ("Hamyar"), Incoming Mechanical Engineering Cohort Sharif University of Technology
Department mentoring programme in which senior students support new students under their faculty advisor *2025 – 2026*

- Responsible for the 18 students advised by Prof. Farzam Farahmand, in a cohort of 120; also supported the others.
- Acted as the link between the faculty advisor and the students, carrying out the tasks he assigned; helped new students settle into university life, answered their questions and followed up on their academic needs.
- Encouraged healthy relationships among classmates and helped run the department's orientation programme.

Mehvar Scientific Association Sharif University of Technology
Student scientific association of the Department of Mechanical Engineering *2024 – 2026*

- Member of the General Council, 33rd term (2025 – 2026).
- Executive team for the department's Yalda Night (Dec 2024) and Nowruz (Mar 2025) events.

Honors

- Ranked 590th among 147,317 participants (top 0.4%), Iranian National University Entrance Exam (Konkur), 2023.